

Chapter 3

Production and Cost and Theory of the Firm

Production: In economics, production is the process of transforming inputs (like labor, capital, and land) to outputs such as goods and services. Production means creating the utility of a commodity.

Production Function: Production function means the technical relationship between inputs and outputs in the process of production.

The standard equation for a production function is expressed as $Q = f(L, K)$.

Where,

Q = Total output

f = Functional relationship

L = Labor

K = Capital

In terms of time or period, production function can be divided into two categories: (a) short-run production function, (b) long-run production function.

(a) Short-run production function: In the short-run production function at least one factor of production remains fixed.

$Q = f(L, K^0)$, where K is fixed.

(b) Long-run production: In the long-run production all factors of production are variable.

$Q = f(L, K)$. Where both labor (L) and capital (K) change.

Short-run vs long-run production function:

SR	LR
1. In the short-run production function at least one factor of production remains fixed.	1. In the long-run production all factors of production are variable.
2. Limited flexibility	2. High flexibility
3. $Q = f(L, K^0)$	3. $Q = f(L, K)$
4. Factor ratio changes	4. Factor ratio doesn't change
5. Follow law of diminishing returns	5. Follow law of returns to scale.

Factors of production:

- Land: Natural resources such as surface, mineral, air, rivers, sea etc.
- Labor: Mental or physical effort done by a man with the view of producing output.

- Capital: Man-made goods used in the production process
- Organization: Entrepreneur or coordinator of all factors of production.

Cost of production: Cost of production means the total amount a firm spends to make goods and services.

Cost function: The cost of production is a mathematical formula that shows the total amount needed to produce a specific output. It is written as

$$C = f(Q)$$

Where, total cost (C) depends directly on the quantity (Q) of goods produced.

Short-run cost function: In short-run, cost function is indicated by fixed cost and variable cost.

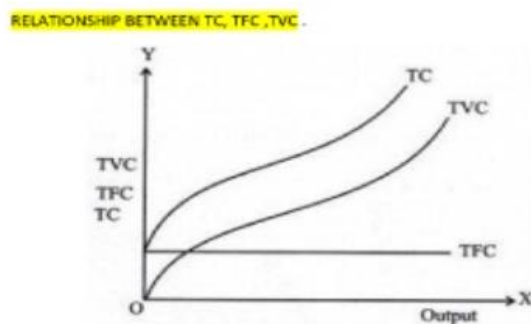
$$C = f(Q) + b; \text{ where } C = \text{Cost}, f(Q) = \text{variable cost}, b = \text{fixed cost}$$

Long-run cost function: In the long run, all costs of production are variable.

$$C = f(Q); \text{ where } C = \text{Cost}, f(Q) = \text{variable cost}$$

Total Cost (TC): The total expenditure incurred by an organization on the factors of production which are required for the production of a commodity is known as Total Cost. In simple terms, total cost is the sum of total fixed cost and total variable cost at different output levels.

$$TC = TFC + TVC$$



Total Fixed cost: All the costs which are not related to the quantity of production, i.e., the costs which do not change with the quantity of production, increase or decrease, but remain constant, are called fixed costs. Generally, house rent, rent, interest on long-term loans, salaries of senior officials, etc. are included in fixed costs.

Total Variable Cost: The cost that changes as a result of a change in the production volume of an organization is called variable cost. If the production volume increases the amount of variable cost increases and if the production volume decreases, this type of cost decreases. For example, wages of workers, cost of purchasing raw materials, electricity bills, etc. are included in variable costs.

Average Cost (AC): The average cost is obtained by dividing the total cost of production by the quantity.

$$\text{Average Cost (AC)} = \frac{\text{Total Cost (TC)}}{\text{Quantity (Q)}}$$

Marginal Cost (MC): The additional cost added to the total cost of producing one additional unit of a particular good is called marginal cost.

$$\text{Marginal Cost (MC)} = \frac{\text{Change in Total Cost } (\Delta TC)}{\text{Change in Quantity } (\Delta Q)}$$

Average Fixed Cost (AFC): Total fixed cost (TFC) is divided by the quantity of production to get average fixed cost (AFC). That is, AFC.

$$\text{Average Fixed Cost (AFC)} = \frac{\text{Total Fixed Cost (TFC)}}{\text{Quantity (Q)}}$$

Average Variable Cost (AVC): Total variable cost (TVC) divided by the quantity of production is the average variable cost (AVC)

$$\text{Average Variable Cost (AVC)} = \frac{\text{Total Variable Cost (TVC)}}{\text{Quantity (Q)}}$$

Relationship between AC and MC:

Average cost is the ratio of total production costs to the total quantity produced.

$$\text{Average Cost (AC)} = \frac{\text{Total Cost (TC)}}{\text{Quantity (Q)}}$$

On the other hand, the amount by which total cost increases to produce one additional unit is marginal cost (MC).

$$\text{Marginal Cost (MC)} = \frac{\text{Change in Total Cost } (\Delta TC)}{\text{Change in Quantity } (\Delta Q)}$$

There is a close relationship between average cost (AC) and marginal cost (MC). This is explained in three stages as follows:

- In the first stage, the initial stage of production, AC and MC usually decrease with the increase in production due to internal and external cost adjustments. However, the rate

at which AC decreases is greater than MC. Therefore, at this stage, average cost (AC) is greater than marginal cost (MC), i.e., $AC > MC$.

- In the second stage, when average cost (AC) reaches its minimum at the desired level of production, average cost and marginal cost are equal to each other. That is, $AC = MC$
- In the third stage, after production reaches the minimum level, if production is increased further, average and marginal cost continue to increase. However, the growth rate of marginal cost is higher than average cost. Therefore, after the transition, average cost is less than marginal cost, i.e., $AC < MC$.

The relationship between AC and MC is explained in the diagram.

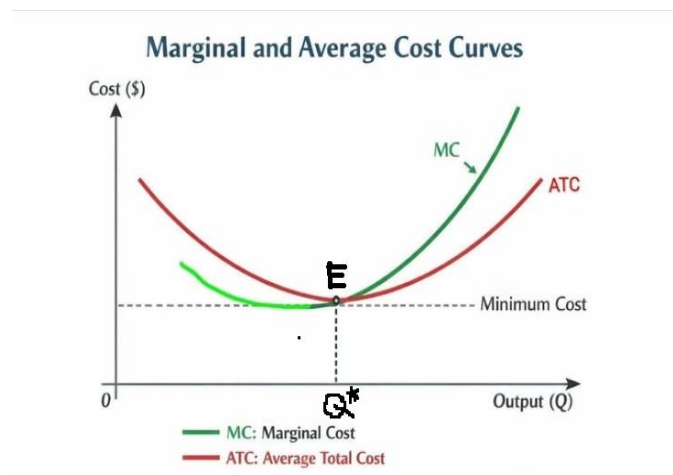


Figure: AC and MC

In the figure, AC and MC are the average and marginal cost curves respectively. Until the quantity OQ is produced, the AC line gradually slopes downwards to the right. As long as the AC line slopes downwards, the MC line remains below the AC line, i.e., at this stage, the quantity of production $AC > MC$ is OQ. At this OQ production level, at point E, i.e., at the lowest point of the AC line, the MC line crosses the AC line from below and rises upwards. Therefore, at the lowest point of the AC line, $MC = AC$ has become. After the production of OQ, both AC and MC slope upwards to the right and the MC line is above the AC line because the rate of increase in marginal cost is higher than the average cost. That is, after the production of OQ, $MC > AC$ will be. The AC line slopes downwards to point E and then upwards. E is the lowest point of the AC line. Therefore, since it slopes downwards first and then upwards, the shape of the AC and MC lines is like the English letter "U".

Using the following table compute the Total Cost (TC), Marginal Cost (MC), Average Cost (AC), Average Variable Cost (AVC), Average Fixed Cost (AFC).

Q	FC	VC	TC	AC	MC	AFC	AVC
0	100	0					
1	100	60					
2	100	80					
3	100	95					
4	100	100					
5	100	105					
6	100	120					
7	100	140					
8	100	170					
9	100	200					
10	100	210					